

[nmap]

nmap -sn [IP] - 가장 기본 스캔

```
(zizon@zizon)-[~]  
$ nmap -sn 172.16.29.11  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-24 19:56 KST  
Nmap scan report for 172.16.29.11  
Host is up (0.00070s latency).  
MAC Address: 00:0C:29:ED:03:6B (VMware)  
Nmap done: 1 IP address (1 host up) scanned in 0.15 seconds
```

호스트 살아있는지 확인

→ **ping sweep**(핑 스캔). **OS**는 못잡음. 살아있는지 여부만.

nmap -sV [IP] - 서비스 스캔(가장 많이 씀)

```
(zizon@zizon)-[~]  
$ nmap -sV 172.16.29.11  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-24 19:57 KST  
Nmap scan report for 172.16.29.11  
Host is up (0.00016s latency).  
Not shown: 994 closed tcp ports (reset)  
PORT      STATE SERVICE VERSION  
21/tcp    open  ftp      vsftpd 3.0.5  
22/tcp    open  ssh      OpenSSH 8.7 (protocol 2.0)  
53/tcp    open  domain   ISC BIND 9.16.23 (RedHat Linux)  
80/tcp    open  http     Apache httpd 2.4.62 ((Rocky Linux))  
3306/tcp  open  mysql    MariaDB 5.5.5-10.5.29  
8000/tcp  open  http     Tornado httpd 6.5.1  
MAC Address: 00:0C:29:ED:03:6B (VMware)  
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 7.26 seconds
```

→ 열려있는 포트 + 서비스 버전

SSH인지 **SFTP**인지, **Apache**인지 **Nginx**인지, **DB** 버전까지 다 나옴.

nmap -O [IP] - 운영체제(OS) 탐지

```
(zizon@zizon)-[~]
$ nmap -O 172.16.29.11
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-24 19:58 KST
Nmap scan report for 172.16.29.11
Host is up (0.00056s latency).
Not shown: 994 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
53/tcp    open  domain
80/tcp    open  http
3306/tcp  open  mysql
8000/tcp  open  http-alt
MAC Address: 00:0C:29:ED:03:6B (VMware)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.19
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.79 seconds
```

→ OS guess (정확도는 환경 따라 다름)

※ -sn 과는 별개. -sn에는 OS 정보 포함 X.

실무에서 '정확한 OS 확인'은 보통 Banner나 HTTP Header로 더 확인함.

nmap -p- [IP] - 포트 전체 스캔

```
(zizon@zizon)-[~]
$ nmap -p- 172.16.29.11
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-24 20:01 KST
Nmap scan report for 172.16.29.11
Host is up (0.00042s latency).
Not shown: 65529 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
53/tcp    open  domain
80/tcp    open  http
3306/tcp  open  mysql
8000/tcp  open  http-alt
MAC Address: 00:0C:29:ED:03:6B (VMware)

Nmap done: 1 IP address (1 host up) scanned in 2.63 seconds
```

→ 0~65535 전체 스캔

웹서버 숨겨진 포트 찾을 때 필수.

nmap -sV -sC -O -p- [IP] - 강력 조합(실무에서 제일 자주 씬)

```
(zizon@zizon)-[~]
$ nmap -sV -sC -O -p- 172.16.29.11
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-24 20:03 KST
Nmap scan report for 172.16.29.11
Host is up (0.00080s latency).
Not shown: 65529 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.5
22/tcp    open  ssh      OpenSSH 8.7 (protocol 2.0)
|_ ssh-hostkey:
|_ 256 9b:5c:db:d0:44:c3:cb:3e:ad:f5:98:4b:ac:3a:e8:3c (ECDSA)
|_ 256 a8:ed:13:f8:62:c5:82:c8:ac:40:ed:a3:5e:01:f9:a4 (ED25519)
53/tcp    open  domain   ISC BIND 9.16.23 (RedHat Linux)
|_ dns-nsid:
|_ bind.version: 9.16.23-RH
80/tcp    open  http     Apache httpd 2.4.62 ((Rocky Linux))
|_ http-server-header: Apache/2.4.62 (Rocky Linux)
|_ http-title: PHP 8.4.14 - phpinfo()
3306/tcp  open  mysql    MariaDB 5.5.5-10.5.29
|_ mysql-info:
|_ Protocol: 10
|_ Version: 5.5.5-10.5.29-MariaDB
|_ Thread ID: 1669
|_ Capabilities flags: 63486
|_ Some Capabilities: Support41Auth, SupportsCompression, ODBCClient, IgnoreSigpipes, Speaks41P
rotocolOld, ConnectWithDatabase, InteractiveClient, IgnoreSpaceBeforeParenthesis, LongColumnFlag
, DontAllowDatabaseTableColumn, Speaks41ProtocolNew, SupportsTransactions, SupportsLoadDataLocal
, FoundRows, SupportsAuthPlugins, SupportsMultipleStatements, SupportsMultipleResults
|_ Status: Autocommit
|_ Salt: 1}E!q;qFHyH:+fBVk9wL
|_ Auth Plugin Name: mysql_native_password
8000/tcp  open  http     Tornado httpd 6.5.1
|_ http-server-header: TornadoServer/6.5.1
|_ http-title: Jupyter Server
|_ Requested resource was /login?next=%2Ftree%3F
MAC Address: 00:0C:29:ED:03:6B (VMware)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4)
Network Distance: 1 hop
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/subm
it/ .
Nmap done: 1 IP address (1 host up) scanned in 18.74 seconds
```

-sV : 버전

-sC : 기본 **NSE** 스크립트

-O : OS

-p- : 전체 포트

→ "이 서버가 뭐하는 애냐?" 한방에 끝.

nmap -p 22,80,443 [IP] - 특정 포트만 스캔

```
(zizon@zizon)-[~]  
$ nmap -p 22,80,443 172.16.29.11  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-24 20:04 KST  
Nmap scan report for 172.16.29.11  
Host is up (0.00033s latency).  
  
PORT      STATE SERVICE  
22/tcp    open  ssh  
80/tcp    open  http  
443/tcp   closed https  
MAC Address: 00:0C:29:ED:03:6B (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.20 seconds
```

nmap -sS [IP] - 스텔스 스캔(SYS Scan)

```
(zizon@zizon)-[~]  
$ nmap -sS 172.16.29.11  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-24 20:07 KST  
Nmap scan report for 172.16.29.11  
Host is up (0.000088s latency).  
Not shown: 994 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
53/tcp    open  domain  
80/tcp    open  http  
3306/tcp  open  mysql  
8000/tcp  open  http-alt  
MAC Address: 00:0C:29:ED:03:6B (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.35 seconds
```

→ 방화벽 우회 확률 증가. 실무 레드팀 기본.

nmap -f [IP] - Firewall 우회 (느리지만 은밀)

```
(zizon@zizon)-[~]  
$ nmap -f 172.16.29.11  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-24 20:08 KST  
Nmap scan report for 172.16.29.11  
Host is up (0.00011s latency).  
Not shown: 994 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
53/tcp    open  domain  
80/tcp    open  http  
3306/tcp  open  mysql  
8000/tcp  open  http-alt  
MAC Address: 00:0C:29:ED:03:6B (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.33 seconds
```

→ 패킷을 쪼개 전송 (**fragment**)

단, 실무에서 드물게만 사용.

nmap 172.16.29.0/24 - 호스트 여러 개 스캔

```
(zizon@zizon)-[~]
$ nmap 172.16.29.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-24 20:09 KST
Nmap scan report for 172.16.29.1
Host is up (0.00036s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
3389/tcp   open  ms-wbt-server
6000/tcp   open  X11
MAC Address: 70:85:C2:72:C5:98 (ASRock Incorporation)

Nmap scan report for 172.16.29.11
Host is up (0.00011s latency).
Not shown: 994 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
53/tcp    open  domain
80/tcp    open  http
3306/tcp  open  mysql
8000/tcp  open  http-alt
MAC Address: 00:0C:29:ED:03:6B (VMware)

Nmap scan report for 172.16.29.20
Host is up (0.000094s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
81/tcp    open  hosts2-ns
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
2049/tcp  open  nfs
8000/tcp  open  http-alt
MAC Address: 00:0C:29:C4:0B:5D (VMware)

Nmap scan report for 172.16.29.33
Host is up (0.0044s latency).
Not shown: 993 closed tcp ports (reset)
PORT      STATE SERVICE
53/tcp    open  domain
```

nmap -T4 [IP]

```
(zizon@zizon)-[~]  
$ nmap -T4 172.16.29.11  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-24 20:12 KST  
Nmap scan report for 172.16.29.11  
Host is up (0.00012s latency).  
Not shown: 994 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
53/tcp    open  domain  
80/tcp    open  http  
3306/tcp  open  mysql  
8000/tcp  open  http-alt  
MAC Address: 00:0C:29:ED:03:6B (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.29 seconds
```

빠르게

Nmap의 핵심 중 하나.

웹, ssh, smb, ftp 등 서비스별 취약점 점검 스크립트 실행.

예)

nainx

코드 복사

”

```
nmap --script http-enum 172.16.29.11
```

nginx

코드 복사

```
nmap --script vuln 172.16.29.11
```

→ 자동 취약점 스캔 느낌.

14) 웹 서버 스캔에 유용한 옵션

nginx

 코드 복사

```
nmap --script http-title 172.16.29.11
```

→ 웹 페이지 제목 확인

nginx

 코드 복사

```
nmap --script http-headers 172.16.29.11
```

→ Apache/Nginx 버전 노출 여부 확인

nginx

 코드 복사

```
nmap --script http-methods 172.16.29.11
```

→ PUT/DELETE 허용 여부 검사 (중요)

실무에서 **nmap**을 어떻게 쓰냐?

1) 침투테스트 / 레드팀

- 대상 서버 **IP** 범위 확인 (**-sn**)
- 서비스 식별 → 공격 벡터 결정 (**-sV**)
- 버전 기반 취약점 조사
- **SSH? MySQL? FTP?**
→ **Hydra**로 브루트포스 들어갈지 결정
- 웹이면 브프랑 연계
→ **nmap**으로 포트 확인 → **burp**로 인젝션/**XSS** 테스트

2) 블루팀

- 불필요하게 열린 포트 발견
- 잘못된 서비스 버전 확인
- 방화벽 정책 점검
- 서버가 공격받고 있는 흔적 탐지(스캔 로그)

3) 네트워크/시스템 운영

- 어떤 서버가 살아있는지 주기적 확인
- 포트 오픈 상태 점검
- 장비 교체 시 확인
- 서비스 죽었는지 체크 자동화